

Betti Mathematics:

Ontological Compression through

Recursive Symbolic Codex

A Speculative Theoretical Framework

Gregory Betti

Founder, Betti Labs

GitHub: <https://github.com/Betti-Labs>

August 2025

IMPORTANT ACADEMIC DISCLAIMER

This project presents a speculative theoretical mathematical framework. "Betti Mathematics" is a proposed mathematical system that extends beyond established mathematical foundations. While grounded in legitimate academic research and existing mathematical precedents, this framework represents theoretical exploration rather than empirically validated mathematics. Academic Positioning: This work follows precedents in theoretical physics where speculative mathematical frameworks are developed to explore potential new understanding. All theoretical constructs presented herein require extensive validation and should be understood as proposed rather than established mathematics. Research Status: This framework is in active research and development phase. All mathematical constructs, algorithms, and theoretical propositions should be considered as hypothetical pending rigorous peer review and empirical validation.

Table of Contents

- 0. Introduction (README)
- 1. Chapter 0: Preface
- 2. Chapter 1: Collapse-Expansion Dynamics
- 3. Chapter 2: Recursive Symbolic Codex
- 4. Chapter 3: Ontological Structures
- 5. Chapter 4: Compression Algorithms
- 6. Chapter 5: Mathematical Foundations
- 7. Chapter 6: Theoretical Applications
- 8. Chapter 7: Validation Methods
- 9. Chapter 8: Advanced Topics
- 10. Chapter 9: Future Directions

Complete Textbook Content

This PDF contains the complete Betti Mathematics theoretical framework. The full textbook includes detailed mathematical formulations, theoretical proofs, implementation examples, and validation methodologies across all chapters. For the complete LaTeX source and detailed mathematical notation, please refer to the LaTeX files in the `textbook/latex/` directory of this package. All visual assets, code implementations, and supporting materials are included in the complete BettiMathematics package.